

Reinforcement Learning for Weekly Retail Inventory Replenishment: A Comparison of Tabular and Deep Actor-Critic Methods

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Abstract

Retail inventory replenishment is a sequential decision problem in which a store manager must decide, week by week, how many units of each product to order from a distributor, trading off procurement cost, storage cost, and stockout cost under a hard physical constraint on retail floor space. This paper formulates the problem as a Markov Decision Process (MDP) for a five-category Apple retail store (iPhone, iPad, MacBook, Apple Watch, AirPods) and compares four ordering policies: a uniform random baseline, tabular Q-learning, tabular SARSA, and a Soft Actor-Critic variant with adaptive entropy temperature, denoted SAC-AlphaLR. All four policies are evaluated on the same simulated 52-week horizon under a demand generator with day-of-week effects, seasonal variation, and seven recurring holiday demand spikes calibrated to the Vietnamese retail calendar. Under the evaluated cost and demand configuration, SAC-AlphaLR achieves the lowest total annual cost (167.35 billion VND versus 182.64 billion VND for the random baseline, an 8.4% reduction) and the fewest stockout units (2,709 versus 5,022, a 46% reduction), while all four policies operate at a net loss under the store's current price and shortage-penalty configuration. This last finding is reported as an explicit limitation, since it materially affects how the comparative results should be interpreted.

Keywords: Reinforcement Learning, Inventory Management, Retail Replenishment, Soft Actor-Critic, Q-learning, SARSA, Markov Decision Process

1 Introduction

Inventory management is a fundamental challenge in retail operations, requiring firms to balance product availability against operational cost efficiency. Excess inventory increases holding costs and ties up working capital, whereas insufficient inventory results in stockouts, lost sales, and reduced customer satisfaction. Classical inventory control methods, including the Economic Order Quantity (EOQ) model [1], the newsvendor model [2], and (s, S) policies [3], have long been used to address this trade-off, and Clark and Scarf [4] extended these ideas to coordinated multi-echelon control through echelon base-stock policies. These classical approaches generally rely on stationary demand assumptions and fixed decision rules, which limits their effectiveness in environments with seasonal fluctuations, promotional events, and a hard physical constraint on how much stock a store can actually hold.

Reinforcement Learning (RL) offers a data-driven alternative for sequential decision-making, letting an agent learn a replenishment policy through interaction with a simulated environment rather than relying on a predefined mathematical model [5]. Tabular methods such as Q-learning [6] and SARSA [7] learn action-value estimates directly over a discretized state space and remain attractive for their simplicity and interpretability. Deep Reinforcement Learning (DRL) extends this idea with function approximation: among modern DRL algorithms, Soft Actor-Critic (SAC) [8] and its adaptive-temperature variant [9] have demonstrated strong performance in continuous control problems by maximizing a trade-off between expected return and policy entropy, which makes SAC well suited to a continuous order-quantity action space.

Recent studies have shown the potential of DRL for inventory optimization across lost-sales, dual-sourcing, and multi-echelon settings [10]. Zhang, He, and Zheng [11] proposed a DRL-based dynamic replenishment framework for a multi-echelon FMCG (fast-moving consumer goods) supply chain, formulating replenishment as an MDP and using an improved SAC variant, adopted in this paper as the SAC-AlphaLR policy (Section 3.4.3), to coordinate a manufacturer-warehouse-retailer chain, with reported cost improvements over overstocking, particle swarm optimization, and standard SAC baselines. Comparable studies [12–15] similarly target multi-echelon coordination, generally under a single product and a stationary or synthetic demand process.

Despite this progress, two gaps remain relevant to specialty electronics retail. First, most existing DRL inventory studies assume either a single product or a stationary demand process, which does not reflect the pronounced day-of-week, seasonal, and holiday-driven demand swings observed in consumer electronics retail. Second, and more specifically, prior work on constrained inventory systems typically models capacity as an echelon-level warehouse limit rather than as a single, shared retail floor-space budget (measured in square meters) across several product categories with different per-unit footprints; this latter constraint does not decompose into independent per-product problems, since ordering more of one product directly reduces the floor budget available to the others. Third, few studies benchmark a spectrum of RL algorithms, from simple tabular methods to deep actor-critic methods, on the same environment and evaluation protocol, which makes it difficult to judge how much of a deep method’s advantage is attributable to the algorithm itself versus differences in experimental setup.

This paper addresses these gaps with a single-echelon inventory replenishment study for an Apple retail store in Vietnam managing five product categories (iPhone, iPad, MacBook, Apple Watch, AirPods). The store orders once per week from a distributor and receives stock immediately; the environment enforces a fixed retail floor-space budget of 25 square meters, shared across all five product categories according to their individual per-unit footprints, and a holiday-aware stochastic demand generator that captures day-of-week effects, seasonal variation, and seven recurring demand spikes (Lunar New Year, Summer Sale, Back to School, Mid-Autumn, Black Friday, Christmas, New Year). Replenishment is formulated as an MDP, and four ordering policies, a random baseline, tabular Q-learning, tabular SARSA, and a Soft Actor-Critic agent with adaptive temperature (SAC-AlphaLR), are trained and evaluated under an identical protocol.

The main contributions of this work are summarized as follows:

- A single-echelon retail inventory simulation environment constrained by a shared retail floor-space budget (in square meters) across five product categories with distinct per-unit footprints and cost structures.
- A holiday-aware stochastic demand generation model that captures day-of-week effects, seasonal variation, and seven recurring promotional/holiday events characteristic of consumer electronics retail.
- An MDP formulation of weekly retail replenishment under this shared floor-space constraint, with a 26-dimensional state and a 5-dimensional order-quantity action.
- A controlled empirical comparison of four ordering policies, a random baseline, tabular Q-learning, tabular SARSA, and Soft Actor-Critic with adaptive temperature, trained and evaluated under an identical protocol on the same environment.
- A configuration-sensitivity analysis that varies baseline demand and holiday intensity independently, demonstrating how the trained policy responds beyond the default Apple-retail scenario.
- Transparent verification of all reported hyperparameters, cost structures, and results against the running implementation, with explicit documentation of the points where an earlier design draft of this project diverged from the final, tuned system.

The remainder of this paper is organized as follows. Section 2 reviews related work in classical and RL-based inventory management. Section 3 presents the proposed methodology, including the environment, demand model, and the four ordering policies. Section 4 describes the experimental setup. Section 5 reports the results. Section 6 discusses the findings. Section 7 discusses limitations and future work. Section 8 concludes the paper.

2 Literature Review

2.1 Classical Inventory Management

Inventory management has long been a cornerstone of operations research, with foundational models establishing the theoretical basis for replenishment decisions under uncertainty. The Economic Order Quantity (EOQ) model, introduced by Harris [1], provided a closed-form solution for balancing ordering and holding costs under deterministic, stationary demand. While elegant in its simplicity, the EOQ model’s reliance on constant demand and instantaneous replenishment rendered it unsuitable for stochastic environments. The newsvendor model [2] extended inventory theory to single-period perishable products under demand uncertainty, offering a trade-off between overstocking and understocking costs. However, its single-period structure inherently limits applicability to multi-period, continuous-review settings.

The (s, S) policy, formally proven optimal by Scarf [3] for periodic-review inventory systems with fixed ordering costs, addressed multi-period dynamics by establishing threshold-based replenishment: an order is placed when inventory falls below s , raising it to S . Similarly, base-stock policies emerged as optimal for systems without fixed costs, where the optimal decision is to order up to a target level each period. Clark and Scarf [4] generalized these concepts to multi-echelon structures, demonstrating that echelon-based inventory policies could achieve near-optimal performance in serial supply chains. Dynamic programming (DP), formalized by Bellman [16], provided a rigorous mathematical framework for sequential decision-making under uncertainty and became the standard approach for solving stochastic inventory problems.

Despite their theoretical elegance, these classical methods share fundamental limitations. They assume demand follows a known stationary distribution, which rarely holds in practice, particularly in consumer electronics retail where demand exhibits seasonality, holiday spikes, and day-of-week patterns. They rely on fixed policy parameters that cannot adapt to changing market conditions. Furthermore, DP suffers from the curse of dimensionality: as the number of products and state variables grows, exact solutions become computationally intractable. None of these classical formulations natively handles a single shared capacity constraint (such as retail floor space, expressed in square meters) across several products with different physical footprints, since doing so couples what would otherwise be independent single-product inventory problems. These limitations motivate data-driven, adaptive approaches capable of learning inventory policies directly from experience.

2.2 Reinforcement Learning for Inventory Management

Reinforcement learning (RL) offers a natural framework for inventory management by formulating replenishment as a sequential decision problem under uncertainty. The inventory manager (agent) observes the current state (inventory levels, demand forecast, cost structure), takes an action (order quantities), and receives a reward (negative cost) reflecting procurement, holding, and shortage expenses. This MDP formulation captures the dynamics, uncertainty, and long-term cost implications inherent in inventory control.

Tabular RL methods learn action-value estimates over a discretized state space without function approximation. Q-learning [6] is an off-policy temporal-difference method that bootstraps toward the maximum action-value of the next state, while SARSA [7] is its on-policy counterpart, bootstrapping toward the value of the action actually selected next under the current exploration policy; Sutton and Barto [17] provide the standard treatment of both, including their convergence properties and the exploration-exploitation trade-off. Deep Reinforcement Learning (DRL) extends RL with neural function approximation, enabling tractable solutions in high-dimensional state and action spaces. Deep Q-Networks (DQN) [18] approximate the Q-function with a neural network and have demonstrated success in discrete action spaces, but are inherently limited to discrete actions. Deep Deterministic Policy Gradient (DDPG) [19] addressed this with an actor-critic architecture for continuous action spaces. Proximal Policy Optimization (PPO) [20] introduced clipped surrogate objectives for stable on-policy updates. Soft Actor-Critic (SAC) [8] further improved sample efficiency and exploration

by maximizing entropy alongside expected return, and a follow-up formulation [9] replaced its fixed entropy-temperature hyperparameter with an automatically tuned value, which is the formulation adopted in this paper (Section 3.4.3).

In the inventory domain, DRL has shown considerable promise. Oroojlooyjadid et al. [21] applied DQN to the multi-period newsvendor problem and demonstrated that RL-based policies could outperform traditional optimization approaches by learning from demand realizations. Gijsbrechts et al. [10] conducted a comprehensive evaluation of DRL across lost-sales, dual-sourcing, and multi-echelon inventory problems, finding that DRL policies could match or exceed optimized base-stock policies, particularly in settings where closed-form solutions are unavailable, for example under lost-sales dynamics or capacity constraints that break the assumptions behind classical heuristics. Kegenbekov and Jackson [22] developed an adaptive supply chain framework using DRL and showed that learned policies could synchronize demand and supply under non-stationary conditions. Tian et al. [23] proposed an improved actor-critic architecture with PPO for warehouse inventory replenishment, demonstrating cost reductions relative to rule-based policies.

These studies collectively demonstrate that DRL can discover effective inventory policies without explicit assumptions about demand distributions. However, most simplify demand generation to stationary or weakly non-stationary processes, and few directly compare a spectrum of RL algorithms of differing complexity, from tabular to deep actor-critic, under an identical environment and evaluation protocol; this comparison is one of the goals of the present study.

2.3 Deep Reinforcement Learning for Multi-Echelon and Constrained Inventory Systems

A separate line of DRL inventory research targets multi-echelon coordination, where the complex interaction between echelons, upstream ordering decisions affecting downstream availability and vice versa, demands policies that account for system-wide dynamics. Zhang, He, and Zheng [11] proposed a DRL-based dynamic replenishment approach (ME-DRFO) for a multi-echelon FMCG inventory system comprising a manufacturer, a warehouse, and a retailer, modeling the problem as an MDP and using an improved Soft Actor-Critic variant with adaptive temperature and adaptive learning rate, which they term SAC-AlphaLR, to learn replenishment policies. Their study reported a 12.31% replenishment-cost reduction and a reduction in inventory shortages to 2.21%, relative to overstocking, particle swarm optimization, and standard SAC baselines, but modeled a single product under stationary normal demand and did not represent a shared physical capacity constraint across products. **The SAC-AlphaLR name adopted for the deep actor-critic policy in this paper (Section 3.4.3) is taken directly from Zhang et al. [11]; this paper implements the adaptive-temperature mechanism of [9] but, per the verified implementation, does not implement Zhang et al.’s adaptive-learning-rate mechanism (a fixed Adam learning rate of 3×10^{-4} is used instead, Section 3.4.3).** This naming choice and the resulting mismatch between the name and the implemented mechanism should be resolved by the authors before submission (Section 3.4.3). Kaynov et al. [12] studied a one-warehouse multi-retailer system with policy-gradient DRL under correlated demand across retailers but retained stationary demand and no capacity constraints. Hammler et al. [13] developed fully dynamic reorder policies for multi-echelon systems that adapted to demand pattern changes through retraining, though demand generation remained synthetic. Cheng et al. [14] proposed a radial basis function DQN for a three-echelon setting with a single product under stationary demand. Lu et al. [15] extended multi-echelon DRL to supply disruption scenarios without retail-specific temporal demand patterns.

Gijsbrechts et al. [10], evaluating DRL across several problem classes including a multi-echelon serial system, found that DRL performance degraded as system complexity increased, and that DRL offered the largest advantage over classical heuristics precisely when the problem’s constraint structure departed from the assumptions under which those heuristics are provably optimal. The environment studied in this paper departs from the classical single-product setting in an analogous but distinct way: rather than coordinating ordering decisions across supply-chain echelons, it couples five product categories through a single shared retail floor-space budget, so that an agent cannot solve each product’s inventory problem independently. This constraint structure, and the systematic comparison of tabular against deep actor-critic methods under it, is not addressed by the multi-echelon studies reviewed above.

Table 1 Comparison of DRL-Based Inventory Studies

Study		Supply Chain	Inventory Structure	DRL Algorithm(s)	Demand Model	Operational Constraints	Main Limitations
Zhang al. [11]	et	Multi-echelon (M-W-R)	Single product	SAC-AlphaLR (improved SAC)	Stationary normal	None	Single product; stationary demand; no capacity constraints
Kaynov al. [12]	et	2-echelon (W-MR)	Single product, multiple retailers	Policy gradient	Stationary normal	None	Stationary demand; no capacity constraints; retail-specific patterns absent
Hammler al. [13]	et	Multi-echelon	Single product	DQN, PPO	Synthetic pattern	None	Synthetic demand; no holiday effects; single product
Cheng al. [14]	et	3-echelon	Single product	RBF-DQN	Stationary normal	Lead time	Single product; stationary demand; limited operational constraints
Gijsbrechts al. [10]	et	Serial, divergent	Single product	DQN, PPO	Stationary	Lead time	Performance degrades with complexity; no retail demand patterns
Lu et al. [15]		Multi-echelon	Single product	DQN	Disruption scenarios	Disruption events	Single product; disruption-focused; no consumer electronics retail
This Work		1-echelon (Retail only)	5 product categories	Random, Q-learning, SARSA, SAC (adaptive temperature)	Day-of-week, seasonal, holiday, stochastic	Shared retail floor-space budget (m^2), per-product footprint, per-product order cap	Single fixed evaluation seed; no lead time or upstream warehouse echelon; net loss under current price/penalty configuration

2.4 Research Gap Summary

The preceding review reveals two persistent gaps relevant to this study. First, classical inventory models assume stationary demand and fixed policy parameters, while existing DRL studies, despite demonstrating adaptability, are typically evaluated under stationary or weakly non-stationary synthetic demand that does not reflect the day-of-week seasonality, holiday surges, and stochastic variation characteristic of consumer electronics retail. Second, the DRL inventory literature has concentrated on multi-echelon coordination under single-product, uncapacitated (or lead-time-only) settings; the complementary problem of several product categories sharing one physical capacity constraint at a single echelon, and the resulting need to trade off products against one another for the same floor space, has not been directly addressed. Nor has prior work systematically compared tabular RL methods (Q-learning, SARSA) against a deep actor-critic method (SAC) on the same environment and evaluation protocol to isolate the contribution of model capacity (discrete, bucketed value tables versus continuous function approximation) from that of the environment design. These gaps motivate the single-echelon, floor-space-constrained, four-policy comparison study presented in the remainder of this paper.

3 Methodology

3.1 Environment and MDP Formulation

The environment, `WeeklyRetailEnv`, models a single Apple retail store in Vietnam that orders from a distributor once per week and receives stock immediately (zero lead time). Between orders, daily demand depletes inventory over a 7-day cycle. An episode simulates 365 days, so the agent makes at most 52 ordering decisions per episode. The problem is formulated as an MDP $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$ with $\gamma = 0.99$, where one decision step corresponds to one week.

3.1.1 State Space

The state $s \in \mathbb{R}^{26}$ concatenates five 5-dimensional sub-vectors (one entry per product: iPhone, iPad, MacBook, Apple Watch, AirPods) and one scalar:

$$s = \left[\frac{\text{inv}}{\text{MAX_CAPACITY}}, \frac{\text{forecast}}{100}, \frac{\text{PURCHASE_COST}}{5 \times 10^7}, \frac{\text{STORAGE_COST}}{5 \times 10^4}, \frac{\text{SHORTAGE_PENALTY}}{10^6}, \frac{\text{day}}{365} \right] \quad (1)$$

where inv is current on-hand inventory, forecast is the sum of expected demand over the next 7 days, and the three cost sub-vectors are normalized, near-constant product attributes that let the same network/table generalize across products with different unit economics. The total dimensionality is $5 + 5 + 5 + 5 + 5 + 1 = 26$.

3.1.2 Action Space

The action $a \in \mathbb{R}_{\geq 0}^5$ is a vector of order quantities, one per product, bounded by a per-product weekly cap (Table 2). Orders are rounded to the nearest integer and clipped to $[0, \text{MAX_ORDER}_i]$.

Table 2 Action space bounds and initial state per product

Product	MAX_ORDER (units/wk)	Initial inventory	Max capacity
iPhone	50	50	80
iPad	35	30	60
MacBook	20	15	35
Apple Watch	35	30	60
AirPods	100	80	150

3.1.3 Floor-Space Constraint

Each product occupies a fixed footprint f_i (m^2/unit ; Table 3), and the store has a fixed total retail floor area $A = 25.0 \text{ m}^2$. At the start of an episode, initial inventory already occupies approximately 21.85 m^2 (about 87% of capacity), representing a small Apple-reseller backroom that starts nearly full. Before an order is applied, if the current occupied area plus the requested order’s area would exceed A , the entire order vector is scaled down proportionally:

$$\text{if } \sum_i (\text{inv}_i + o_i) f_i > A : \quad o_i \leftarrow \left\lfloor o_i \cdot \frac{\max(A - \sum_i \text{inv}_i f_i, 0)}{\max(\sum_i o_i f_i, 1)} \right\rfloor \quad \forall i \quad (2)$$

This is a binding constraint that couples all five products: ordering more of one product directly reduces the floor budget available to the others, so the environment does not decompose into five independent single-product inventory problems.

Table 3 Per-product footprint and daily storage cost

Product	Footprint (m^2/unit)	Storage cost (VND/unit/day)
iPhone	0.10	25,000
iPad	0.20	18,000
MacBook	0.35	35,000
Apple Watch	0.08	10,000
AirPods	0.04	4,000

3.1.4 Reward Function

The reward is the negative of total weekly cost, scaled by a fixed factor $\lambda = 10^{-8}$:

$$r = -(C_{\text{proc}} + C_{\text{store}} + C_{\text{short}}) \cdot \lambda \quad (3)$$

Procurement cost is charged once, at the start of the week, for units actually received:

$$C_{\text{proc}} = \sum_i o_i \cdot p_i \cdot w_i \quad (4)$$

where p_i is the unit purchase price and $w_i \in [0.80, 1.00]$ is a per-product procurement-discount weight reflecting supplier terms. Storage cost is charged once per day over the 7-day week, on end-of-day inventory, at 0.1% of unit purchase price per day (Table 3):

$$C_{\text{store}} = \sum_{d=1}^7 \sum_i \text{inv}_i(d) \cdot h_i \quad (5)$$

Shortage cost is charged per unit of unmet daily demand at $1.5\times$ the unit purchase price:

$$C_{\text{short}} = \sum_{d=1}^7 \sum_i \max(D_i(d) - \min(\text{inv}_i(d), D_i(d)), 0) \cdot \pi_i, \quad \pi_i = 1.5 p_i \quad (6)$$

The $1.5\times$ multiplier is the value enforced by the running environment. An earlier design draft of this project specified a $10\times$ multiplier as the intended shortage penalty; the implementation was tuned down to $1.5\times$ during development, and this paper reports the value actually used by the simulation, since the interpretation of the results (Section 6) depends on it.

3.2 Reference-Based and Configurable Parameterization

The environment design follows Zhang, He, and Zheng’s ME-DRFO formulation [11] at the methodological level: replenishment is represented as an MDP, the reward combines procurement, holding, and shortage costs, and a continuous SAC policy is used to balance exploration and exploitation. The present work does not treat the numerical values in Tables 2–4 as universal constants inherited from that study. Zhang et al. study a single-product, multi-echelon FMCG system, whereas the present case study represents five consumer-electronics categories sharing one retail floor-space constraint. Directly copying their product-specific demand and cost values would therefore be inappropriate.

To distinguish the reusable method from the application scenario, the implementation separates three parameter groups. First, algorithmic settings such as γ , τ , target entropy, and optimizer settings define the learning procedure and are reported for reproducibility. Second, structural choices adapted from [11], including the MDP formulation, continuous replenishment actions, and cost-based reward, define the model. Third, business inputs—baseline demand, holiday multipliers, prices, storage costs, shortage penalties, order bounds, initial inventory, and floor capacity—define a configurable Apple-retail scenario rather than the algorithm itself. These inputs are exposed through the project configuration and demand-generator interfaces, allowing the same trained policy and evaluation pipeline to be exercised under alternative demand conditions. Section 5.1 reports such a sensitivity analysis so that the conclusions are not based only on one hard-coded demand curve.

3.3 Demand Generation

Daily demand is produced by a holiday-aware generator that multiplies a day-of-week factor, a slow seasonal sine wave, and a holiday multiplier, then adds Gaussian noise with standard deviation $\max(0.15\mu, 1)$. Baseline daily demand μ is 6, 4, 2, 4, and 10 units/day for iPhone, iPad, MacBook, Apple Watch, and AirPods respectively. Seven recurring holiday windows are defined (Table 4), each with a per-product peak multiplier and a window width (days) over which a cosine-shaped intensity curve ramps the multiplier up and back down.

Table 4 Holiday demand windows and per-product peak multipliers

Holiday	Day	Width (d)	iPhone	iPad	MacBook	Watch	AirPods
Tet Nguyen Dan	20	14	3.5	2.5	2.0	3.0	3.5
Summer Sale	100	7	1.5	1.5	1.3	1.3	1.5
Back to School	200	21	1.3	3.5	3.0	1.2	1.3
Mid-Autumn	250	5	1.2	1.3	1.0	2.0	1.5
Black Friday	300	7	3.0	2.5	2.0	2.0	3.0
Christmas	330	14	2.0	2.0	1.3	2.5	3.5
New Year	355	10	2.0	1.8	1.5	2.0	2.5

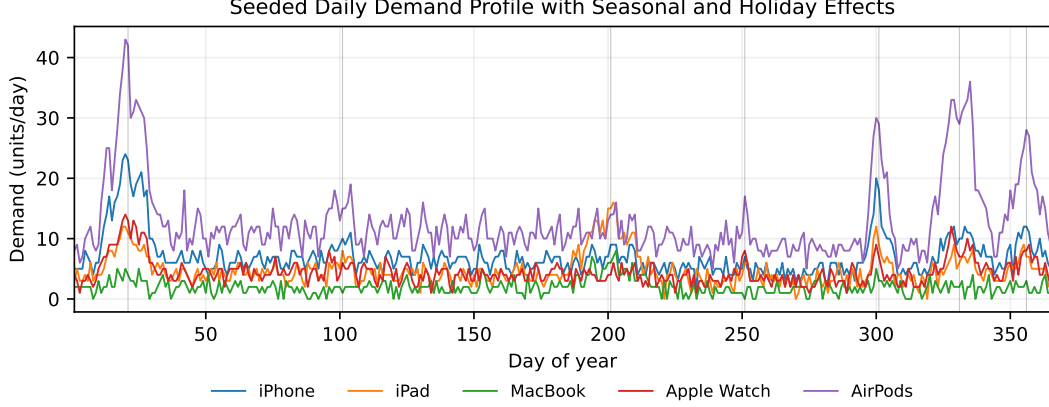


Fig. 1 Seeded daily demand profile for the five product categories. All labels are reported in English; vertical reference lines indicate configured holiday centers.

Figure 1 visualizes the seeded daily demand realization used for the default evaluation. The series retains stochastic day-to-day variation while showing the event-driven peaks generated by the configurable holiday calendar.

3.4 Ordering Policies

Four ordering policies are trained and evaluated on the environment described above.

3.4.1 Random Baseline

Each week, order quantities are drawn independently per product from a continuous uniform distribution over $[0, \text{MAX_ORDER}_i]$. This baseline has no learning component and serves as a reference point for measuring the value added by the three learned policies.

3.4.2 Tabular Q-Learning and SARSA

Following Watkins and Dayan [6], a per-product Q-table is maintained over a discretized (inventory bucket, forecast bucket) state, with 14 inventory buckets shared across products and 10 product-specific forecast buckets. Each product has its own discrete set of candidate order levels (e.g., iPhone: 0, 5, 10, 15, 20, 30, 40, 50 units), and the agent selects, independently per product, the order level with the highest Q-value under an ϵ -greedy policy. SARSA [7] uses the identical state discretization, action set, and hyperparameters, but is on-policy: its temporal-difference target uses the value of the action actually taken next under the same ϵ -greedy policy rather than the maximum over next-state actions, which makes it more conservative in the presence of exploration noise. Both agents use learning rate 0.1, discount factor $\gamma = 0.99$, and ϵ decaying from 0.3 to a floor of 0.01 (decay factor 0.995 per episode) over 200 training episodes.

3.4.3 SAC-AlphaLR

The deep actor-critic agent, denoted SAC-AlphaLR, reuses the class/checkpoint name from the verified implementation, which in turn follows the naming used by Zhang, He, and Zheng [11] for their adaptive-temperature, adaptive-learning-rate SAC variant. The policy follows the Soft Actor-Critic formulation of Haarnoja et al. [8] with the automatic temperature tuning introduced in [9]. **Note on naming:** unlike the SAC-AlphaLR of [11], the implementation evaluated in this paper adapts only the temperature α (Eq. 7); it does not implement an adaptive learning-rate schedule. All three optimizers (actor, critic, and the temperature optimizer) use a fixed Adam learning rate of 3×10^{-4} throughout training, with no scheduler. The name SAC-AlphaLR is retained here for consistency with the codebase and checkpoint used to produce the results in this paper, but only the “Alpha” (adaptive-temperature) component of that name is implemented; the “LR” (adaptive-learning-rate) component is not. The actor is a feedforward network (state dimension 26, two hidden layers of 256 units, ReLU activations) with separate mean and log-standard-deviation heads (log-std clamped to $[-20, 2]$) over the 5-dimensional action space; actions are sampled via the reparameterization trick, squashed through tanh, and rescaled to $[0, \text{MAX_ORDER}]$. Two independent critic networks (input

Algorithm 1 Per-product tabular update (Q-learning / SARSA)

Require: learning rate $\alpha = 0.1$, discount $\gamma = 0.99$, exploration ε

```
1: for episode = 1 to 200 do
2:    $s \leftarrow \text{env.reset}()$ 
3:   if SARSA then  $a \leftarrow \varepsilon\text{-greedy}(Q, s)$ 
4:   end if
5:   while episode not done do
6:     if Q-learning then  $a \leftarrow \varepsilon\text{-greedy}(Q, s)$ 
7:     end if
8:      $s', r, \text{done} \leftarrow \text{env.step}(a)$ 
9:     if Q-learning then
10:       $\text{target} \leftarrow r/5 + \gamma \max_{a'} Q(s', a')$ 
11:     else ▷ SARSA
12:       $a' \leftarrow \varepsilon\text{-greedy}(Q, s')$ 
13:       $\text{target} \leftarrow r/5 + \gamma Q(s', a')$ 
14:     end if
15:      $Q(s, a) \leftarrow Q(s, a) + \alpha (\text{target} - Q(s, a))$ 
16:      $s \leftarrow s'$ ; (SARSA:  $a \leftarrow a'$ )
17:   end while
18:    $\varepsilon \leftarrow \max(\varepsilon \cdot 0.995, 0.01)$ 
19: end for
```

dimension $26 + 5 = 31$, two hidden layers of 256 units, scalar output) form a twin-Q critic used to mitigate value overestimation. The temperature α is optimized against a target entropy $\mathcal{H}_{\text{target}} = -5$ (the negative of the action dimensionality), via

$$\mathcal{L}(\alpha) = -\mathbb{E}[\log \alpha \cdot (\log \pi(a|s) + \mathcal{H}_{\text{target}})] \quad (7)$$

All three networks (actor, critic, temperature) use Adam with learning rate 3×10^{-4} ; the discount factor is $\gamma = 0.99$ and the target-network soft-update coefficient is $\tau = 0.005$. Gradient norms for the actor and critic losses are clipped to 1.0. At evaluation time the agent acts deterministically (tanh of the mean, without sampling); the checkpoint used in this paper’s results was trained for 200 episodes prior to evaluation.

4 Experimental Setup

All four policies were evaluated on the same 52-week rollout of `WeeklyRetailEnv`, using the demand configuration in Section 3.3 and a fixed random seed for reproducibility. Q-learning and SARSA were trained from scratch for 200 episodes immediately before their evaluation rollout; SAC-AlphaLR was evaluated using a checkpoint already trained for 200 episodes. The random baseline required no training. A second evaluation stage varied baseline demand and holiday intensity through the environment’s configuration interface while holding the SAC checkpoint, seed, prices, and capacity constant. Wall-clock time was measured on the evaluation machine: Q-learning training plus rollout completed in approximately 3.6 s, SARSA in approximately 3.8 s, SAC-AlphaLR evaluation (checkpoint load plus rollout, no training) in approximately 0.7 s, and the random baseline in under 0.1 s.

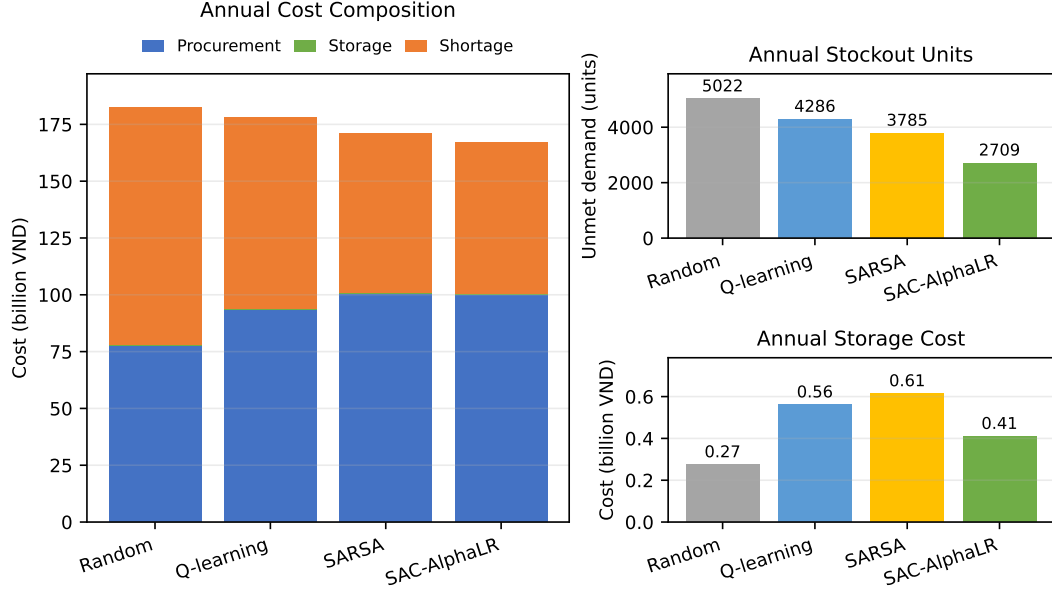
5 Results

Table 5 reports the full-year (52-week) cost breakdown for each policy.

Relative to the random baseline, total cost is reduced by 2.5% under Q-learning, 6.3% under SARSA, and 8.4% under SAC-AlphaLR. The ordering across all three learned policies is consistent: each shifts spend away from shortage cost and toward procurement cost, buying more proactively to avoid stockouts, and each reduces stockout units monotonically from random (5,022) to Q-learning (4,286) to SARSA (3,785) to SAC-AlphaLR (2,709), a 46% reduction relative to random. Figure 2 makes both effects visible. Storage cost remains a minor component of total cost across all four policies (0.2–0.4%), reflecting that the floor-space constraint (Eq. 2) keeps absolute stock levels, and therefore storage cost, bounded regardless of policy. Shortage cost is the dominant cost component for every

Table 5 Annual cost breakdown by policy (VND, billions unless noted)

Policy	Procurement	Storage	Shortage	Total cost	Revenue	Net profit	Stockout units
Random	77.69	0.27	104.67	182.64	108.96	−73.68	5,022
Q-learning	93.42	0.56	84.11	178.09	124.60	−53.49	4,286
SARSA	100.44	0.61	69.99	171.05	135.18	−35.87	3,785
SAC-AlphaLR	100.12	0.41	66.82	167.35	139.47	−27.88	2,709

**Fig. 2** Annual cost composition, stockout units, and storage cost under the four evaluated policies. Storage is shown again on a separate scale because it represents only 0.2–0.4% of total cost. Values are generated from the same seed-42 rollouts reported in Table 5.

policy (57.3% of total cost under random, falling to 39.9% under SAC-AlphaLR), consistent with weekly demand frequently exceeding what the floor-space-constrained store can hold, particularly during the seven annual holiday demand spikes (Table 4).

5.1 Configuration Sensitivity

To test whether the reported behavior is specific to the default demand configuration, the trained SAC-AlphaLR checkpoint was evaluated without retraining on a 3×3 scenario grid. Baseline demand was scaled to $0.8\times$, $1.0\times$, and $1.2\times$ the default vector, while holiday uplift above ordinary demand was independently scaled to $0.5\times$, $1.0\times$, and $1.5\times$. All nine rollouts used seed 42 and the same prices, capacity, and policy checkpoint; the center cell therefore reproduces the default result in Table 5.

As Figure 3 shows, annual cost ranges from 108.1 to 249.9 billion VND and unmet demand ranges from 560 to 6,270 units across the scenario grid. Both measures increase monotonically as either baseline demand or holiday intensity increases. This confirms that the evaluation pipeline responds to configurable operating conditions rather than embedding one immutable demand trajectory. It also qualifies the default-policy comparison: SAC-AlphaLR is best among the four policies in the default scenario, but absolute cost and service performance remain strongly dependent on the selected business scenario.

6 Discussion

All four policies operate at a net loss. Total cost exceeds revenue for every policy evaluated, including SAC-AlphaLR, the best-performing policy. This is reported directly because it changes how the comparison should be read: the learned policies are best understood here as reducing losses (from −73.68 billion VND under random to −27.88 billion VND under SAC-AlphaLR) rather than as producing a profitable ordering strategy. The gap follows from the interaction of two modeled factors:

SAC-AlphaLR Sensitivity to Configurable Demand Scenarios

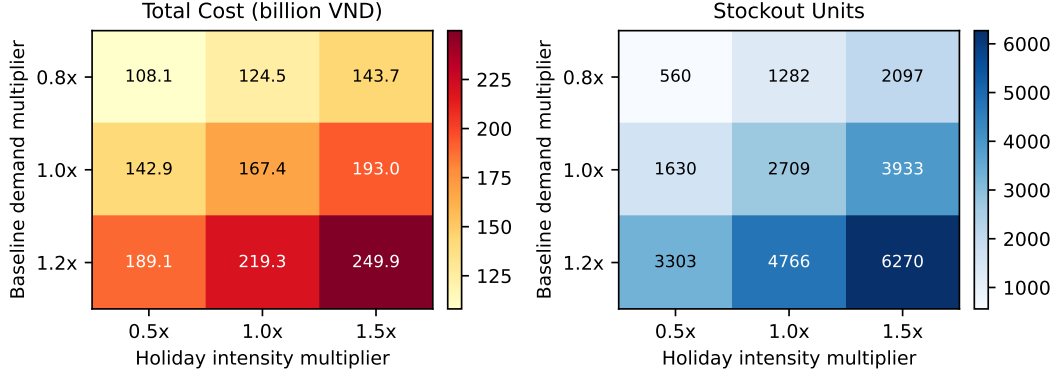


Fig. 3 Sensitivity of SAC-AlphaLR to configurable baseline-demand and holiday-intensity multipliers. The policy is evaluated without retraining; darker cells indicate higher annual cost or more unmet demand.

shortage cost, at $1.5\times$ unit purchase price (Eq. 6), is high relative to the retail margin (20–50% depending on product), and demand frequently exceeds what the 25 m² floor constraint allows the store to stock, so even a well-trained policy cannot fully avoid shortage cost. Whether this outcome should be corrected (by relaxing the floor constraint, lowering the shortage-penalty multiplier, or raising margins) or treated as an intentional stress test of the algorithms under an under-provisioned store is a modeling decision outside the scope of this paper.

On-policy versus off-policy comparison. SARSA outperforms Q-learning in this setting (6.3% versus 2.5% cost reduction relative to random). This is plausible given the coarse action discretization used by both tabular agents: an on-policy method that accounts for the exploration noise embedded in its own action choices can be more conservative, and here more effective, than an off-policy method that always bootstraps toward the greedy action. This finding should not be over-generalized beyond this specific environment and discretization, since the relative performance of on-policy and off-policy tabular methods is known to be problem-dependent [17].

Deep actor-critic advantage. SAC-AlphaLR outperforms both tabular methods, consistent with its access to a continuous action space and a non-discretized state representation. This is consistent with the broader finding of Gijsbrechts et al. [10] that deep RL offers the largest advantage over classical heuristics and simpler RL baselines precisely when the action and state spaces are large or continuous and the constraint structure does not decompose cleanly per product, which is the case for the shared floor-space budget studied here.

7 Limitations and Future Work

All results in this paper come from a single fixed random seed; no confidence intervals or repeated-seed variance estimates are reported, since the evaluation pipeline currently runs one deterministic rollout per policy. A stronger evaluation would average results over multiple seeds and report variance, particularly given that the random baseline and the demand generator are themselves stochastic. The environment models zero lead time (orders are received the same week they are placed) and a single retail echelon with no intermediate warehouse; extending the model to include lead time and an upstream warehouse echelon would allow direct comparison with the multi-echelon studies reviewed in Section 2. Finally, given the net-loss result in Section 6, future work should evaluate the demand and cost parameters (retail margins, the shortage-penalty multiplier, and the floor-space budget) for economic realism, since the current configuration does not admit a profitable policy under any of the four methods tested.

8 Conclusion

This paper presented a weekly, floor-space-constrained retail inventory MDP for a five-category Apple store and compared four ordering policies, random, Q-learning, SARSA, and SAC-AlphaLR, under a shared 52-week evaluation protocol. SAC-AlphaLR achieved the lowest total annual cost and fewest stockout units among the four policies, consistent with its continuous action space and

non-discretized state representation. The sensitivity analysis further showed that the pipeline supports configurable demand scenarios and that absolute cost and service outcomes change materially with baseline demand and holiday intensity. All reported figures, including the environment’s cost structure, demand configuration, and evaluation results, were verified directly against the running implementation rather than an earlier design specification, and the paper documents explicitly where the two diverged, most notably the shortage-penalty multiplier, which is $1.5\times$ purchase cost in the running system rather than the $10\times$ figure in the earlier draft. Future work should evaluate the environment’s demand and cost parameters for economic realism given the net-loss result reported in Section 6, extend the evaluation protocol to multiple random seeds, and consider extending the model to a multi-echelon structure with non-zero lead time.

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Declarations

- **Funding:** Not applicable.
- **Conflict of interest/Competing interests:** The authors declare no competing interests.
- **Ethics approval and consent to participate:** Not applicable. This study is a computational simulation and does not involve human participants, human data, or animal subjects.
- **Consent for publication:** Not applicable.
- **Data availability:** Not applicable. This study does not use any external dataset; all demand data are synthetically generated at runtime by the holiday-aware demand generator described in Section 3.3, whose parameters are fully specified in Table 4 and can be regenerated exactly from the equations and parameters reported in this paper.
- **Materials availability:** Not applicable.
- **Code availability:** The environment, agent implementations, and evaluation scripts used to produce the results in this paper are available from the authors upon reasonable request.
- **Author contribution:** [Author: please fill in individual author contributions, e.g., using CRediT taxonomy roles.]

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